## Assessing Senegal's External Debt Sustainability[1]

1. *External debt sustainability is preserved under the baseline scenario, and is robust to a series of plausible shocks, including a worsening of borrowing terms. There are little risks of unfavorable debt dynamics, especially if the authorities remain committed to a prudent debt management strategy in the post-HIPC period. The updated DSA yields estimates for the key debt indicators that are broadly consistent with those calculated at the time of the HIPC Completion Point*.

2. **The long-term macroeconomic scenario underlying the DSA is more conservative than the baseline of the April 2004 HIPC Completion Point document.** The key assumptions underlying the long-term macroeconomic framework are: (i) real GDP growth averaging 5.2 percent annually, or about ½ percentage point lower than in the HIPC baseline; (ii) export volume growth averaging 4.9 percent per year, compared to 5.6 percent in the HIPC document; (iii) a doubling of the fiscal deficit (averaging 2.8 percent of GDP) against 1.5 percent of GDP for the HIPC scenario; and (iv) larger new borrowing, reflecting the more expansionary fiscal stance.

3. **Under this macroeconomic scenario, Senegal can maintain a sustainable external debt with ratios well below the reference thresholds.** Senegal's key debt ratios would continue to decline over the medium to long term (Table 1a and Figure 1).

4. **Senegal's external debt remains sustainable over the long term, even in the presence of some exogenous shocks.** Stress tests show that a shock to export growth—with exports falling by about 10 percent in 2005 and 2006—would be the main risk. However, the debt indicators would return to a sustainable level after a temporary hike (Figure 1 and Table 1b). Besides, neither lower economic growth (set at 0.6 percent in 2005 and 2006) nor lower nondebt inflows (with current transfers falling to 2 percent of GDP in 2005 and 2006 from 8 percent in 2004) would lead to unsustainable debt dynamics. Likewise, the debt sustainability path is robust to a combination of moderate shocks involving lower economic growth, lower nondebt inflows and a devaluation (Table 1b). It is also shown that a scenario based on key variables set at their historical averages—the historical scenario—would display a sustainable debt path, thereby suggesting that the results of the baseline scenario are robust.

5. **Senegal's external debt sustainability would also be preserved in the event of a measured deterioration of borrowing terms.** A three-percentage point increase in nominal interest rates for new borrowing—implying a grant element falling to below 20 percent—would lead to a significant deterioration of the debt indicators (Table 1b and Figure 2), but they remain below the indicative thresholds over the long term.

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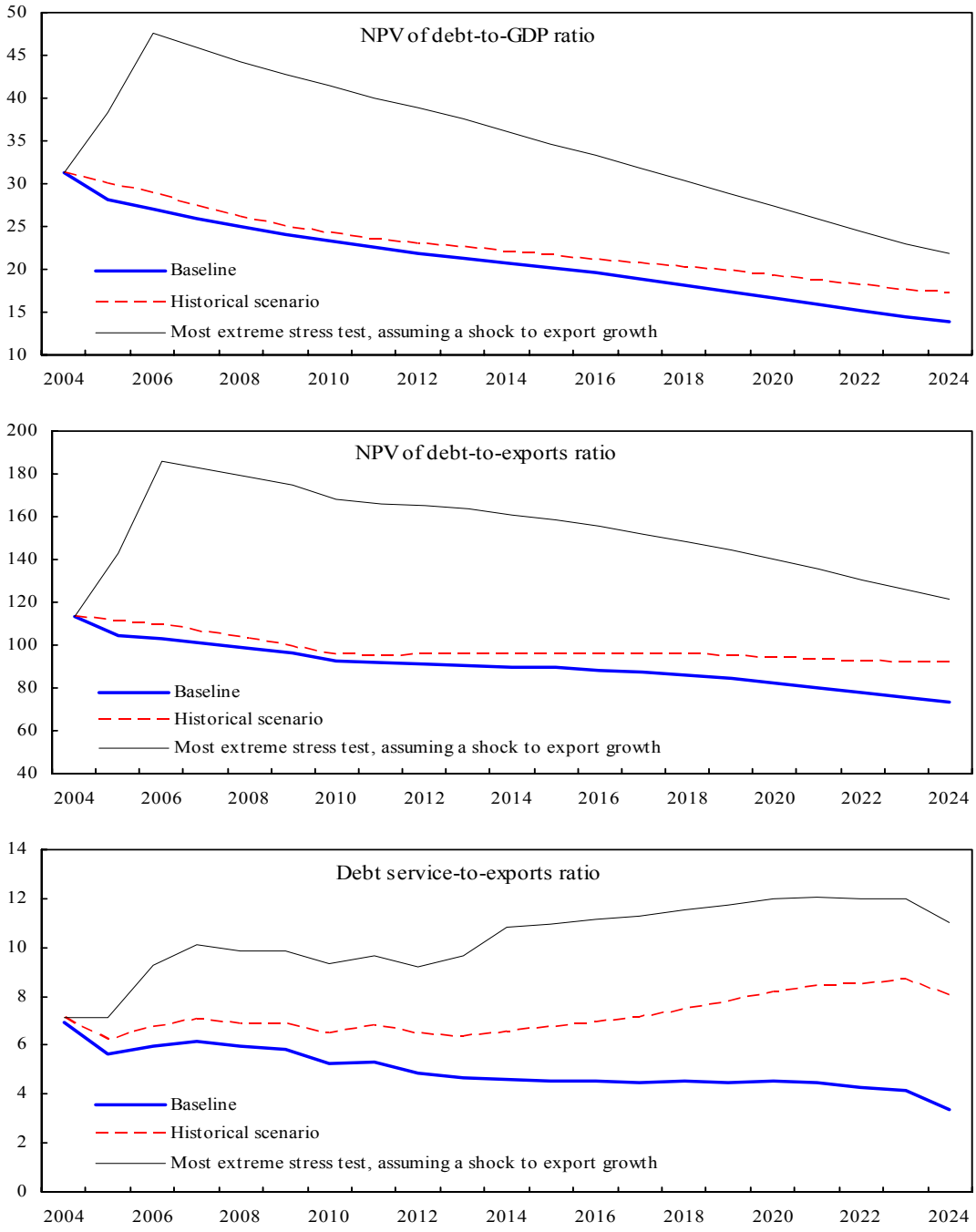
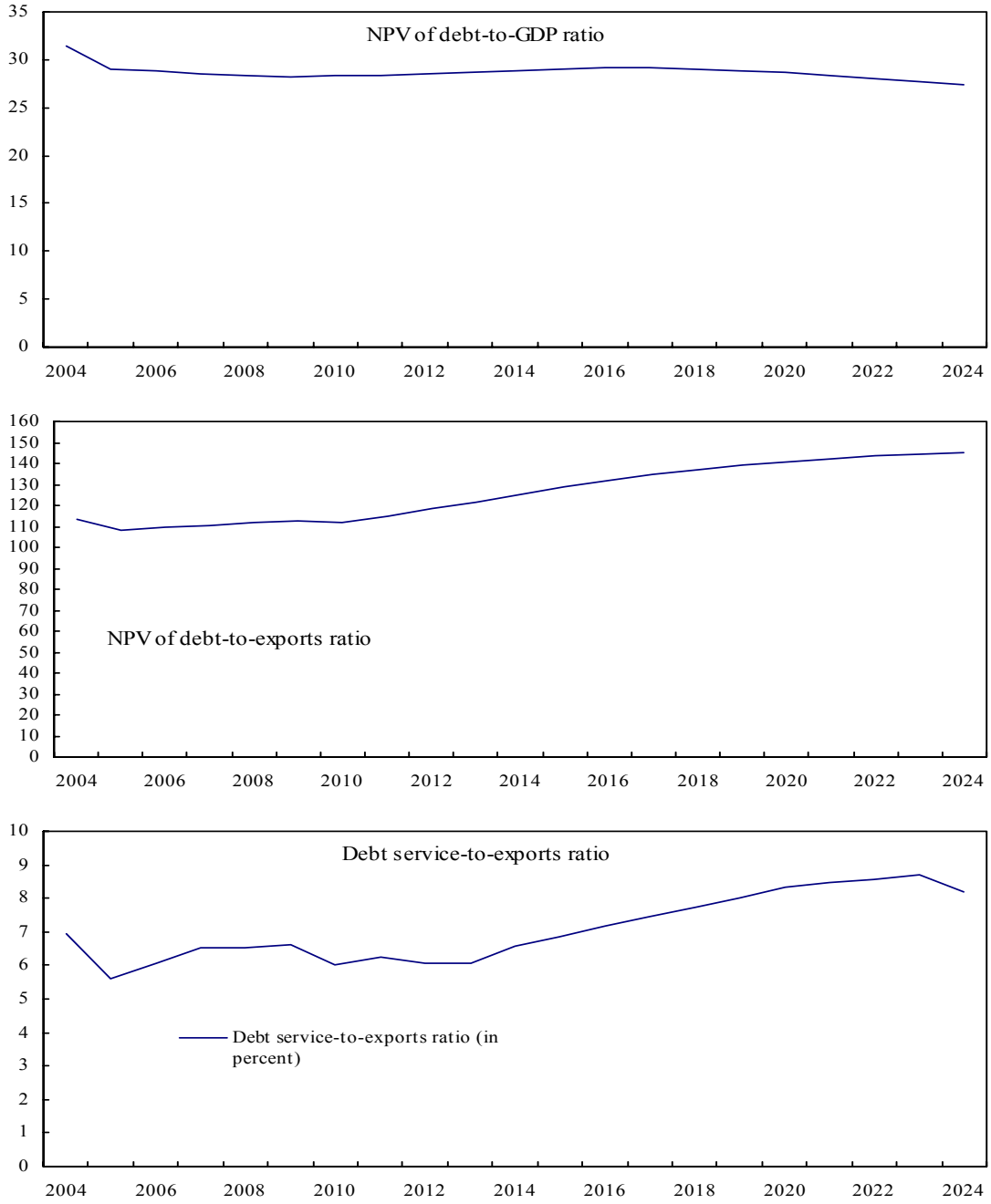
[1] The analysis is based on the DSA template for low-income countries. Senegal's debt sustainability is reviewed in relation to the following reference thresholds: (i) 150 percent for NPV of debt to exports; (ii) 40 percent for NPV of debt to GDP; and (iii) 20 percent for the debt service to exports ratio.

Figure 1. Senegal: Indicators of Public and Publicly Guaranteed External Debt Under Alternative Scenarios, 2004-2024




Source: Staff projections and simulations.

Figure 2. Senegal: Alternative Scenario 2 - Less favorable financing terms 2004-2024
(In percent)




Source: Staff projections and simulations.

Table 1a. Senegal: External Debt Sustainability Framework, Baseline Scenario, 2000-2024 1/
(In percent of GDP, unless otherwise indicated)

| | Actual | | | | Historical Average 6/ | Standard Deviation 6/ | Est. | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | 2000 | 2001 | 2002 | 2003 | | | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | 2024 | 2010-2024 Average |
| **External debt (nominal) 1/** | **69.5** | **67.3** | **66.1** | **54.5** | | | **41.4** | **38.3** | **35.4** | **33.0** | **31.1** | **29.6** | **28.6** | **22.9** | 26.1 |
| o/w public and publicly guaranteed (PPG) | 69.5 | 67.3 | 66.1 | 54.5 | | | 41.4 | 38.3 | 35.4 | 33.0 | 31.1 | 29.6 | 28.6 | 22.9 | 26.1 |
| Change in external debt | -4.7 | -2.3 | -1.2 | -11.6 | | | -13.1 | -3.1 | -2.9 | -2.4 | -1.9 | -1.5 | -1.0 | -0.5 | -0.4 |
| Identified net debt-creating flows | 10.1 | 2.7 | -0.8 | -8.9 | | | 2.2 | 1.7 | 1.7 | 2.3 | 2.3 | 2.3 | 1.9 | 1.1 | 1.5 |
| **Non-interest current account deficit** | **3.5** | **3.3** | **4.9** | **5.4** | **3.6** | **1.7** | **5.4** | **4.9** | **4.6** | **4.9** | **4.9** | **4.8** | **4.2** | **2.9** | 3.5 |
| Deficit in balance of goods and services | 8.4 | 9.7 | 11.2 | 13.0 | | | 12.4 | 11.2 | 11.0 | 11.1 | 10.8 | 10.5 | 9.7 | 5.8 | 7.8 |
| Exports | 30.5 | 30.7 | 30.6 | 28.5 | | | 27.6 | 26.9 | 26.3 | 25.8 | 25.4 | 25.1 | 25.3 | 18.8 | 21.8 |
| Imports | 38.8 | 40.4 | 41.7 | 41.5 | | | 40.1 | 38.1 | 37.2 | 36.9 | 36.2 | 35.5 | 35.0 | 24.6 | 29.5 |
| Net current transfers (negative = inflow) | -5.1 | -6.7 | -7.5 | -8.4 | -4.9 | 2.2 | -8.1 | -7.5 | -7.4 | -7.1 | -6.9 | -6.7 | -6.5 | -4.0 | -5.2 |
| Other current account flows (negative = net inflow) | 0.3 | 0.3 | 1.3 | 0.8 | | | 1.1 | 1.2 | 1.1 | 1.0 | 1.0 | 1.0 | 0.9 | 1.1 | 0.9 |
| **Net FDI (negative = inflow)** | **-1.4** | **-0.9** | **-0.9** | **-0.8** | **-0.9** | **1.4** | **-1.2** | **-1.4** | **-1.5** | **-1.5** | **-1.4** | **-1.4** | **-1.4** | **-0.8** | -1.1 |
| **Endogenous debt dynamics 2/** | **7.9** | **0.3** | **-4.8** | **-13.5** | | | **-2.0** | **-1.8** | **-1.4** | **-1.2** | **-1.1** | **-1.0** | **-0.9** | **-1.0** | -0.9 |
| Contribution from nominal interest rate | 1.6 | 1.4 | 1.0 | 1.2 | | | 0.8 | 0.5 | 0.4 | 0.4 | 0.4 | 0.4 | 0.4 | 0.2 | 0.3 |
| Contribution from real GDP growth | -2.4 | -3.2 | -0.7 | -3.4 | | | -2.8 | -2.3 | -1.8 | -1.7 | -1.5 | -1.4 | -1.4 | -1.2 | -1.3 |
| Contribution from price and exchange rate changes | 8.8 | 2.2 | -5.1 | -11.3 | | | … | … | … | … | … | … | … | … | |
| **Residual (3-4) 3/** | **-14.8** | **-5.0** | **-0.4** | **-2.7** | | | **-15.3** | **-4.8** | **-4.7** | **-4.7** | **-4.3** | **-3.8** | **-2.9** | **-1.6** | -1.9 |
| o/w exceptional financing | -69.0 | -418.5 | -360.1 | -620.5 | | | -1467.6 | -1204.3 | -946.1 | -842.6 | -791.9 | -733.8 | -672.8 | -209.3 | -380.6 |
| NPV of external debt 4/ | … | … | … | 35.5 | | | 31.4 | 28.2 | 27.1 | 26.0 | 25.0 | 24.1 | 23.3 | 13.8 | 18.7 |
| In percent of exports | … | … | … | 124.7 | | | 113.6 | 104.7 | 103.1 | 100.6 | 98.5 | 96.2 | 92.4 | 73.5 | 85.3 |
| **NPV of PPG external debt** | **…** | **…** | **…** | **35.5** | | | **31.4** | **28.2** | **27.1** | **26.0** | **25.0** | **24.1** | **23.3** | **13.8** | 18.7 |
| In percent of exports | … | … | … | 124.7 | | | 113.6 | 104.7 | 103.1 | 100.6 | 98.5 | 96.2 | 92.4 | 73.5 | 85.3 |
| Debt service-to-exports ratio (in percent) | 12.7 | 11.5 | 11.2 | 10.2 | | | 6.9 | 5.6 | 6.0 | 6.2 | 5.9 | 5.8 | 5.2 | 3.4 | 4.5 |
| **PPG debt service-to-exports ratio (in percent)** | 12.7 | 11.5 | 11.2 | 10.2 | | | 6.9 | 5.6 | 6.0 | 6.2 | 5.9 | 5.8 | 5.2 | 3.4 | 4.5 |
| Total gross financing need (billions of U.S. dollars) | 0.3 | 0.3 | 0.4 | 0.5 | | | 0.5 | 0.4 | 0.4 | 0.5 | 0.5 | 0.6 | 0.5 | 0.9 | 0.7 |
| Non-interest current account deficit that stabilizes debt ratio | 8.2 | 5.5 | 6.1 | 17.0 | | | 18.5 | 8.0 | 7.5 | 7.4 | 6.8 | 6.2 | 5.2 | 3.4 | 3.9 |
| **Key macroeconomic assumptions** | | | | | | | | | | | | | | | |
| Real GDP growth (in percent) | 3.0 | 4.7 | 1.1 | 6.5 | 4.1 | 3.3 | 6.0 | 6.4 | 5.2 | 5.0 | 4.9 | 5.0 | 4.9 | 5.3 | 5.1 |
| GDP deflator in US dollar terms (change in percent) | -10.5 | -3.0 | 8.3 | 20.7 | 2.4 | 15.8 | 11.9 | 8.3 | 2.0 | 2.0 | 2.0 | 2.0 | 2.1 | 2.3 | 2.1 |
| Effective interest rate (percent) 5/ | 2.0 | 2.0 | 1.6 | 2.3 | 2.5 | 0.7 | 1.7 | 1.4 | 1.2 | 1.3 | 1.3 | 1.4 | 1.5 | 0.8 | 1.3 |
| Growth of exports of G&S (US dollar terms, in percent) | -7.8 | 2.3 | 8.9 | 19.9 | 3.4 | 12.5 | 15.0 | 12.1 | 4.8 | 5.3 | 5.3 | 5.7 | 7.9 | 5.8 | 5.4 |
| Growth of imports of G&S (US dollar terms, in percent) | -3.3 | 5.6 | 13.1 | 27.8 | 4.5 | 64.8 | 14.5 | 9.6 | 4.8 | 6.1 | 4.9 | 5.3 | 5.5 | 5.0 | 4.8 |
| Grant element of new public sector borrowing (in percent) | … | … | … | … | … | … | 44.3 | 51.4 | 49.6 | 49.9 | 49.4 | 49.2 | 49.1 | 49.8 | 49.5 |
| *Memorandym item:* | | | | | | | | | | | | | | | |
| Nominal GDP (billions of US dollars) | 4.5 | 4.6 | 5.0 | 6.4 | | | 7.6 | 8.8 | 9.4 | 10.1 | 10.8 | 11.6 | 12.4 | 33.7 | 21.3 |

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as $[r - g - \rho(1+g)]/(1+g+\rho+g\rho)$ times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and $\rho$ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability, except for the GDP deflator in dollar terms, for which the average was calculated from 1996 in order to remove the bias introduced by the large 1994 devaluation.

Table 1b. Senegal: Sensitivity Analyses for Key Indicators of Public and Publicly Guaranteed External Debt, 2004-24
(In percent)

| | Est. | Projections | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | 2004 | 2005 | 2006 | 2007 | 2008 | 2009 | 2010 | **2014** | 2024 |
| **NPV of debt-to-GDP ratio** | | | | | | | | | |
| **Baseline** | 31 | 28 | 27 | 26 | 25 | 24 | 23 | **21** | 14 |
| **A. Alternative Scenarios** | | | | | | | | | |
| A1. Key variables at their historical averages in 2004-23 1/ | 31 | 30 | 29 | 27 | 26 | 25 | 24 | **22** | 17 |
| A2. New public sector loans on less favorable terms in 2004-23 2/ | 31 | 29 | 29 | 28 | 28 | 28 | 28 | **29** | 27 |
| **B. Bound Tests** | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2004-05 | 31 | 30 | 30 | 29 | 28 | 27 | 26 | **23** | 15 |
| B2. Export value growth at historical average minus one standard deviation in 2004-05 3/ | 31 | 31 | 34 | 33 | 31 | 30 | 29 | **26** | 16 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2004-05 | 31 | 35 | 40 | 38 | 37 | 35 | 34 | **31** | 20 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2004-05 4/ | 31 | 32 | 34 | 32 | 31 | 30 | 29 | **26** | 16 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 31 | 38 | 48 | 46 | 44 | 43 | 41 | **36** | 22 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/ | 31 | 39 | 37 | 36 | 35 | 33 | 32 | **29** | 19 |
| **NPV of debt-to-exports ratio** | | | | | | | | | |
| **Baseline** | 114 | 105 | 103 | 101 | 99 | 96 | 92 | **90** | 73.5 |
| **A. Alternative Scenarios** | | | | | | | | | |
| A1. Key variables at their historical averages in 2004-23 1/ | 114 | 111 | 110 | 106 | 103 | 100 | 96 | **96** | 92 |
| A2. New public sector loans on less favorable terms in 2004-23 2/ | 114 | 108 | 110 | 110 | 112 | 113 | 112 | **125** | 145 |
| **B. Bound Tests** | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2004-05 | 114 | 105 | 103 | 101 | 99 | 96 | 92 | **90** | 73 |
| B2. Export value growth at historical average minus one standard deviation in 2004-05 3/ | 114 | 143 | 186 | 182 | 178 | 175 | 168 | **161** | 122 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2004-05 | 114 | 105 | 103 | 101 | 99 | 96 | 92 | **90** | 73 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2004-05 4/ | 114 | 117 | 129 | 126 | 123 | 121 | 116 | **111** | 84 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 114 | 139 | 172 | 169 | 165 | 162 | 156 | **149** | 110 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/ | 114 | 105 | 103 | 101 | 99 | 96 | 92 | **90** | 73 |
| **Debt service ratio** | | | | | | | | | |
| **Baseline** | 7 | 6 | 6 | 6 | 6 | 6 | 5 | **5** | 3 |
| **A. Alternative Scenarios** | | | | | | | | | |
| A1. Key variables at their historical averages in 2004-23 1/ | 7 | 6 | 7 | 7 | 7 | 7 | 7 | **7** | 8 |
| A2. New public sector loans on less favorable terms in 2004-23 2/ | 7 | 6 | 6 | 7 | 7 | 7 | 7 | **8** | 12 |
| **B. Bound Tests** | | | | | | | | | |
| B1. Real GDP growth at historical average minus one standard deviation in 2004-05 | 7 | 6 | 6 | 6 | 6 | 6 | 6 | **6** | 7 |
| B2. Export value growth at historical average minus one standard deviation in 2004-05 3/ | 7 | 7 | 9 | 10 | 10 | 10 | 9 | **11** | 11 |
| B3. US dollar GDP deflator at historical average minus one standard deviation in 2004-05 | 7 | 6 | 6 | 6 | 6 | 6 | 6 | **6** | 7 |
| B4. Net non-debt creating flows at historical average minus one standard deviation in 2004-05 4/ | 7 | 6 | 6 | 7 | 7 | 7 | 6 | **7** | 8 |
| B5. Combination of B1-B4 using one-half standard deviation shocks | 7 | 7 | 8 | 9 | 9 | 9 | 8 | **10** | 10 |
| B6. One-time 30 percent nominal depreciation relative to the baseline in 2004 5/ | 7 | 6 | 6 | 6 | 6 | 6 | 6 | **6** | 7 |
| *Memorandum item:* | | | | | | | | | |
| Grant element assumed on residual financing (i.e., financing required above baseline) 6/ | 48 | 48 | 48 | 48 | 48 | 48 | 48 | **48** | 48 |

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 3 percentage points higher than in the baseline, while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.